

Medhansh Garg

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EDUCATION

University of Illinois, Urbana-Champaign

Aug. 2023 – May 2027 (Expected)

- B.S. in Computer Engineering | Relevant Coursework: Data Structures & Algorithms, Computer Systems Engineering, Computer Architecture, Operating Systems, Databases

SKILLS

- **Languages:** TypeScript, Python, Java, C, C++, SQL
- **Frameworks & Web:** React, Next.js, Node.js, FastAPI, Flask, Spring Boot, Prisma, SQLAlchemy, REST APIs
- **Data & Cloud:** PostgreSQL, SQLite, Supabase, Docker, Kubernetes, AWS, Azure, GitHub Actions CI/CD, Vercel, nginx, Git, Linux
- **ML & CV:** PyTorch, MediaPipe, OpenCV, YOLOv8, NumPy, pandas

EXPERIENCE

Hopcharge | Software Engineering Intern

Jun. 2026 – Aug. 2026

- Built the **Ad Engine**, a marketing pipeline (**Next.js 16**, **React 19**, **Prisma/Postgres**) that generates, reviews, and publishes ad creatives to **Meta** and **YouTube**, then learns from performance. Scaled output from 2–3 ads/week to **3/day**, about **8×**
- Made every vendor (LLM, image/video, publishers) swappable through one config variable. Jobs run on a **pg-boss** queue with a serverless **Vercel Cron** fallback, and a feedback loop tracks Meta CPL and YouTube engagement separately to guide new ideas
- Built **VOLT.CV**, a **FastAPI** HR dashboard that automates hiring end-to-end: resume parsing with a 3-tier OCR fallback (**pdfplumber**, **PyMuPDF**, **Tesseract**, plus **spaCy** for field extraction), Gmail reply classification, and one-click bulk interview scheduling
- Secured it with database-level **Row-Level Security** tied to Google identities in **Supabase/PostgreSQL**, and shipped a cross-platform desktop build with **PyInstaller** and **GitHub Actions** CI, targeting 4 OS/architectures
- Built **Grid Ops**, replacing 5 hand-run scripts with a **FastAPI/React (Vite/TS)** service covering demand mapping, churn, forecasting, and pod allocation across **3 Delhi-NCR metros** (about 60 hexes), deployed behind **nginx** on an ARM VM

CS 124 Honors | Lead Web Developer

Aug. 2025 – Present

- Built and lead the CS 124 platform (**Next.js**, **Supabase**), serving **300+ students** and **30+ staff** each semester. Replaced an insecure login prototype with **CILogon** SSO and a 6-role least-privilege system granting temporary role-view access without permanent escalation
- Routed all public pages through a server-side API layer so no DB keys reach the client. Gated dev-to-main deploys on a **Vitest** and **Playwright** E2E suite (isolated test tables, real JWT sessions) in **GitHub Actions** CI
- Automated student/staff, grading, and event management, including batch grading, CSV import, and sprint tracking, cutting manual overhead and errors. Lead an **11-person** dev team through weekly meetings and PR reviews

Care Health Insurance | Cybersecurity Intern

Jun. 2025 – Aug. 2025

- Ran external recon that uncovered **60+ unauthenticated internet-exposed services**. Reverse-engineered **WAF request-signing** into a **Burp Suite** replay PoC, and bypassed **SSL pinning** and root detection using **Frida** and **JADX**

IoT++ | Software Engineering Intern

Jun. 2024 – Aug. 2024

- Built a real-time fire/smoke detection pipeline with **YOLOv8**, improving accuracy **62%** and throughput **13×** through **pruning** and **tiling**. Deployed **NPU-optimized** inference on Orange Pi, containerized with **Docker** and **Kubernetes** on **Azure**

PROJECTS

ASL Adaptive Tutor

github.com/HackOverflow404/SignLanguageLearning

- Built a real-time computer-vision tutor for American Sign Language. Trained a multi-task **triplet-embedding** grader (**PyTorch**) over **MediaPipe** pose landmarks, reaching **81.2% top-1** and **96.5% top-5** accuracy across 60 signs, **2.5×** a DTW baseline
- Benchmarked 4 pose backends across 1,874 clips, then built a live-webcam grader that gives per-parameter feedback (handshape, movement, location) alongside an **SM2 spaced-repetition** scheduler with contrastive minimal-pair drills

AlberFlowy

github.com/HackOverflow404/AlberFlowy

- Reverse-engineered WorkFlowy's private API to support full tree CRUD through a **Node.js CLI**, and built a **C++** Albert plugin with caching, async sync, and **Google OAuth2** automation
- Benchmarked on a 150+ item production vault: live plugin queries return in **0.7ms** (p50) and **1.7ms** (p95) at a **100%** cache hit rate. Caching cuts repeat syncs from about 265ms to **under 1ms**, and the sync engine parses 10k-node trees in about 3ms

RemoteCam

github.com/HackOverflow404/RemoteWebcam

- Built a cross-platform **WebRTC** webcam streaming system with **Firebase** signaling, plus a **QtPython** Linux client and a **Next.js** mobile **PWA** for real-time camera sharing

LEADERSHIP & INVOLVEMENT

Illinois MicroTech | External Vice President | EOH Project Lead

Oct. 2025 – Present

- As *External VP and EOH 2026 Project Lead*, led the Engineering Open House showcase for **20,000+ attendees**